Project 7: AI Voice Agent for Support and Inquiry Handling

Northwind Academy course-inquiry agent · Prepared 18 October 2026 · Stack: Vapi, Deepgram Aura-2 voice, Soniox transcription, GPT-4.1-mini, n8n Cloud, Google Sheets

1. Summary

The project is a voice agent, “Sam”, that answers course inquiries for a synthetic business, Northwind Academy. It looks up fees, refunds, discounts and the next start date live during the call through two n8n-hosted tools. When it cannot answer, it collects the caller’s name and email for admissions follow-up. Every finished call is logged to Google Sheets through an n8n webhook.

Headline results

- Voice latency fell from about 436 ms (Day 1 baseline) to about 190 ms after switching to Deepgram Aura-2.
- LLM first-token latency fell from 629 ms to about 496 ms on average.
- When no tool is used, the caller waits about 750 ms between finishing a sentence and hearing the agent (target: under 800 ms).
- The main bottleneck is the tool call: about 2.06 s per lookup through n8n Cloud. A spoken “Let me check that for you” fills that gap.
- The ten-call test found three failure modes, mainly the agent inventing facts when a tool returned nothing. Prompt fixes (v3, Appendix A) are written and need a re-test.

2. Day 1: Voice agent fundamentals

2.1 Pipeline

Caller audio goes to streaming speech-to-text (Soniox), then to the LLM (GPT-4.1-mini with the system prompt and tools), then to streaming text-to-speech (Deepgram Aura-2), and back to the caller. Two supporting parts shape how natural it feels: endpointing, which decides when the caller has finished speaking, and barge-in, which lets the caller interrupt. Vapi orchestrates all of it. The agent was created in the Vapi dashboard and tested in the browser with Talk to Assistant.

2.2 Voice provider comparison

All three voices ran the same assistant, LLM, transcriber and test questions. Latency is time to first audio taken from Vapi call logs.

	Deepgram Aura-2	ElevenLabs (eleven_turbo_v2)	Vapi built-in voice
Median latency, 13 Oct test	250 ms (n=21)	456 ms (n=8)	423 ms (n=20)
Best / worst	106 / 367 ms	184 / 684 ms	256 / 730 ms
Later calls, 17–18 Oct	166 ms (n=17), 181 ms (n=8)	not re-tested	not re-tested
Naturalness (tester rating, 1–5)	5 (highest)	3 (medium)	3 (medium)
Cost	Measured: about $0.055–0.063 per call minute all-in; TTS was only 7–10% of each bill	Published price roughly $0.05 per 1,000 characters (not measured here)	Not captured

Write-up. I compared three text-to-speech voices on the same Northwind Academy agent. Deepgram Aura-2 was the fastest, with a median of 250 ms in the first test and about 170 ms in later calls, and the tester rated it the most natural of the three. ElevenLabs and Vapi's built-in voice were slower (about 450 ms and 420 ms) and both rated medium. Cost was not a deciding factor: in measured calls, text-to-speech made up only 7–10% of the bill, while the Vapi platform fee made up over 80%. Text-to-speech was never the main delay; even the slowest voice added under half a second per turn. For a course-inquiry agent that needs speed more than emotional range, I chose Aura-2. ElevenLabs would be worth testing again on its newer Flash model if brand voice mattered more than speed.

Caveats. Samples are small (8–21 per voice). The ElevenLabs run used an older model. Per-voice cost was only captured for Aura-2 calls, so the ElevenLabs figure is a published price.

3. Day 2: Conversation design

3.1 Conversation flow

1. **Greeting:** one short line, no menu.

2. **Detect intent:** if unclear, ask one short question.
3. **Branch A, FAQ question:** answer from the tools or the FAQ in one or two sentences.
4. **Branch B, wants to enroll:** collect name, then email, one at a time; repeat the email back; promise a payment link within one business day.
5. **Branch C, out of scope, upset or asks for a person:** one empathetic sentence if upset, then take name and email for admissions follow-up.
6. **Anything else?** A yes returns to step 2; a no leads to a short goodbye and the end-call tool.

(Insert the flowchart image from the chat here.)

3.2 Prompt iteration

- **v1:** short style rules, strict "FAQ only" rule, contact collection, end-call tool.
- **v2:** added a persona ("Sam"), a 1–2 sentence limit, varied sentence openers, explicit interruption rules and a flow section.
- **v3 (Appendix A):** added a Tools section, a fallback rule for failed tools, a rule that the agent cannot see caller accounts, and a limit on repeated follow-up offers.

3.3 Interruption handling

The first test failed because the agent kept talking. The fixes were testing with a headset, asking for a longer answer to interrupt, and checking the Stop Speaking Plan (defaults: 0 words, 0.2 s voice, 1 s backoff). After the fix the agent stopped and recovered:

- Caller: "Wait. Say the… data analytics bootcamp." Agent stopped and answered the bootcamp price.
- Caller: "Wait, wait." Agent: "No problem. Take your time. What else would you like to know?"

4. Day 3: n8n automation

Workflow "Vapi Call Logger": Webhook (POST) → IF (message type equals end-of-call-report) → Google Sheets, Append Row. Vapi's Server URL points at the n8n production webhook, and only the end-of-call-report server message is enabled.

Sheet columns: time, call_id, duration, ended_reason, summary, transcript.

Tested: real browser calls produced rows in the sheet and successful runs in n8n Executions. (Insert screenshots of the workflow canvas, the sheet rows and the Executions list.)

Issue found and fixed: the newer n8n has a Publish button instead of an Active toggle. The production URL only runs after the workflow is published, and edits need re-publishing.

5. Day 4: Function calling

Two Vapi function tools point at a second n8n workflow, “Vapi Tools” (Webhook → Code → Respond to Webhook):

- **lookup_faq(question):** returns course, fee, refund, payment and discount information.
- **check_next_batch:** calculates the next first-Monday start date and the enrollment deadline (3 days earlier) live from today’s date.

Evidence from call logs: three successful lookups of 2,070 ms, 2,081 ms and 2,022 ms. The agent answered correctly from the results, for example the bootcamp at twelve hundred dollars for 12 weeks and the 7-day refund policy.

Problems found and fixed:

- The tool pointed at the Day 3 logger URL, so it returned an empty reply. Fixed by giving each tool the Vapi Tools URL.
- The Respond setting and the Publish step were wrong at first, so edits did not reach the live URL.
- The keyword matching missed phrasings and answered only one part of a two-part question. Fixed by returning the full FAQ text.

Failure handling: each tool has a spoken “Let me check that for you” at request start and a failure message, and the prompt says not to guess when a tool fails. In the failure test, the agent said “Sorry, I couldn’t check that just now” and did not go silent. The caller hung up before the agent could take an email, so this counts as a partial pass.

6. Day 5: Testing and optimization

6.1 Latency report

Stage	Day 1 baseline (13 Oct, 20 turns)	After (17–18 Oct)	Change
Voice, first audio	436 ms avg	about 190 ms avg (n=25)	about −56%
LLM, first token	629 ms avg	496 ms avg, 515 ms median (n=15)	about −21%

Stage	Day 1 baseline (13 Oct, 20 turns)	After (17–18 Oct)	Change
Tool call (n8n)	none	about 2,060 ms avg (n=3)	new cost
Endpointing	54 ms	about 2 ms	negligible
Caller's last word to agent's first audio, no tool	not measured	627–832 ms, median about 750 ms (n=5)	near the 800 ms target
Same, with a tool call	not measured	3,230 ms (n=1)	the main delay
Transcriber, as Vapi reports it	1,496 ms avg	1,404 ms median (n=5)	see note

Bottleneck. My earlier reading blamed transcription. That figure is measured from the moment the caller starts speaking, so it includes their own speaking time, and one 8.5 s outlier came from a noise event, not a real delay. The real bottleneck is the tool call: a turn with a tool took about 3.2 s against about 0.75 s without one. The n8n workflow itself takes 1.7–1.8 s for three trivial nodes, which is platform overhead.

Reductions applied: Aura-2 voice, language fixed to English, short replies, a spoken filler while tools run.

Options still open: put the most common prices into the prompt so most questions skip the tool (hybrid), or move the tool endpoint to a faster server.

Cost: about $0.055–0.063 per minute all-in across three calls ($0.03–0.11 per call). The Vapi platform fee was 80–83%, text-to-speech 7–10%, transcription about 7% and the LLM about 2–3%.

6.2 Ten-call test log

#	Test	Result	Notes
1	SQL course price	Pass	Via tool (self-reported)
2	Next batch start	Pass	check_next_batch, 2,070 ms in the log
3	Courses and installments in one question	**Fail**	First run answered only one part. On retest the tool returned no result and the agent then wrongly said installments were not offered

#	Test	Result	Notes
4	Interruption	Pass after fix	Initially did not stop; passes after the Stop Speaking Plan check and a headset
5	Unclear intent	Pass	Self-reported
6	Enrollment	Pass	Self-reported
7	Out of scope (“Do you teach cooking?”)	Pass	Self-reported; logged turn took 1,268 ms
8	Job guarantee	Pass	Self-reported
9	Refund after 3 weeks	Pass	Refund answer verified in logs: 7 days, none after
10	Tool down	Partial pass	Fallback spoken, no silence; caller hung up before the email step
Extra	Caller says they paid a $100 deposit	**Fail**	Agent invented a partial-payment policy

6.3 Top three failure modes and fixes

1. **Invents facts when a tool returns nothing.** After the failed lookup the agent said “We don’t have installment payments right now”, which contradicts the FAQ. *Fix:* if a tool fails or returns nothing, do not answer that question from memory; say you could not check, try once more, then take name and email. Never say something is unavailable unless the FAQ or a tool says so.
2. **Invents answers about the caller’s own account.** When the caller mentioned a $100 deposit, the agent confirmed a partial-payment policy that does not exist. *Fix:* the agent cannot see accounts, payments or bookings; it says so and offers an admissions follow-up, and never confirms the caller’s claims.
3. **Repetitive follow-up offers.** In several replies the agent ended with a question about the next start date. *Fix:* offer a follow-up at most once per call.

Root cause to check for failure 1: the first lookup in that call returned no result. Open n8n Executions for that time; a red node will show whether the Code node errored or the workflow was mid-publish.

Re-test (not yet run): repeat calls 3, 9 and 10 and the deposit scenario on the v3 prompt, and record the transcripts here.

7. Limitations

- Sample sizes are small (5–25 measurements per metric), so the figures are indicative.
- Several test results are self-reported and are marked as such.
- The v3 prompt fixes are written but not yet re-tested.
- Per-voice cost was measured only for Aura-2.
- Some oddities in assistant lines in the transcripts (such as “I thought”) appear to be speech-to-text artifacts, because assistant text is also transcribed from audio.

Appendix A: Final system prompt (v3)

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[Identity]
You are Sam, the inbound phone assistant for Northwind Academy, an online
school for practical data skills. You sound warm, relaxed, and approachable,
like a friendly front-desk person, not scripted or corporate.

[Style]
- This is a phone call. Keep replies to 1-2 short sentences, never more than
3.
- Answer directly. Don't repeat the question or over-explain.
- Use contractions. Start each reply differently, never the same starter
twice in a row.
- No bullet points, lists, or 'firstly/secondly'.
- Say prices, dates, and times the way you'd say them aloud, like 'twelve
hundred dollars'.
- If asked several things at once, answer them briefly in one reply (max 3
sentences).
- Offer a follow-up question at most once per call. Never end every reply
with a question.
- Ask for one contact detail at a time.

[Tools]
- For questions about courses, prices, refunds, payment options, or
discounts, call lookup_faq first and answer from its result.
- For the next start date or enrollment deadline, call check_next_batch.
Never work out dates yourself.
- Say the result naturally in 1-2 sentences. Never mention tools, servers, or
lookups.
- If a tool fails, times out, or returns nothing: do NOT answer that question
from memory and do not guess. Say you couldn't check that right now, offer to
try once more, and if it fails again take their name and email for admissions
follow-up.
- For anything else (hours, schedule, deferral, certificates, career support,
requirements, enrolling), use the FAQ below.
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[Knowledge Rules]
- Use only tool results and the FAQ below. No outside knowledge, assumptions, or guessing.
- Never say something is unavailable, not offered, or not allowed unless the FAQ or a tool result says so.
- You cannot see caller accounts, payments, deposits, or bookings. If the caller mentions one, say you can't see account details and offer to pass it to admissions with their name and email. Never confirm or agree with claims about their account.
- If the answer isn't available, say you'll pass it to admissions, then collect their name and email. Tell them admissions will follow up within one business day.
- Never claim you sent a message or notified admissions. The caller's details are captured from the call for follow-up.
- Never promise jobs or guaranteed outcomes.
- If asked whether you're an AI, say yes, simply.

[Flow]
1. Greet warmly and ask how you can help.
2. Work out whether they have a question, want to enroll, or need something else. If unclear, ask ONE short question.
3. Questions: use the tools or the FAQ as described above.
4. Enrolling: offer to take their name, then their email, so admissions can send a payment link within one business day.
5. Not in the FAQ, a complaint, or a request for a person: take their name and email for admissions follow-up.
6. Before ending, ask if they need anything else. If not, finish a short goodbye sentence, then use the end-call tool.

[Interruptions and Emotions]
- If the caller interrupts, stop and answer what they just said. Don't return to your previous point.
- If they say 'wait' or 'sorry', say 'No problem,' and listen.
- If you only caught part of what they said, ask them to repeat just that part.
- If they're upset, acknowledge it in one short sentence, then help or move to follow-up.
- If they decline to share details, accept it politely.

[Contact Collection]
- Ask for the name first, then the email. Repeat the email back to confirm before moving on.
- If the email is unclear or incomplete, ask the caller to spell it slowly. Never guess or fill in missing parts.
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[FAQ]
ABOUT: Northwind Academy is an online school teaching practical data skills.
All classes are live online and recorded. Support hours are Monday to Friday,
9 AM to 6 PM Pakistan time.
SCHEDULE: Live classes run Monday, Wednesday, and Friday from 7 PM to 9 PM
Pakistan time. Sessions are recorded and available for 6 months.
DEFERRAL: A student can move to the next month's batch once, free of charge,
if requested at least 2 days before the start date.
CERTIFICATES: Students who attend at least 80 percent of classes and complete
the final project receive a certificate of completion.
CAREER SUPPORT: Data Analytics Bootcamp students get a CV review and one mock
interview. Northwind Academy does not guarantee job placement.
REQUIREMENTS: Students need a laptop with a stable internet connection. All
software used is free.
ENROLLMENT: Callers can enroll on the website or give their name and email so
admissions can send a payment link within one business day.
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Appendix B: Screenshots to insert before submitting

- Vapi assistant page and one successful call
- Flowchart (from the chat)
- n8n canvases: Vapi Call Logger and Vapi Tools
- Google Sheet with at least two call rows
- n8n Executions list showing successful runs
- Transcripts: a live tool lookup, the failure fallback, and an interruption
- One screen recording of a full call that includes a tool lookup